In [8]:

```python
import pandas as pd
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"

print("Original columns:", df.columns.tolist())

# Select correct columns explicitly
df_clean = df[[
    'Country name',
    'Ladder score',
    'Logged GDP per capita',
    'Social support',
    'Healthy life expectancy',
    'Freedom to make life choices',
    'Generosity',
    'Perceptions of corruption'
]].copy()

# Rename properly
df_clean.columns = [
    'Country',
    'Happiness_Score',
    'GDP',
    'Social_Support',
    'Life_Expectancy',
    'Freedom',
    'Generosity',
    'Corruption'
]

print(df_clean.head())
```

```
Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'uppe
rwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expect
ancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder
score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'E
xplained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Expl
ained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']
       Country  Happiness_Score     GDP  Social_Support  Life_Expectancy  \
0      Finland            7.804  10.792           0.969           71.150   
1      Denmark            7.586  10.962           0.954           71.250   
2      Iceland            7.530  10.896           0.983           72.050   
3       Israel            7.473  10.639           0.943           72.697   
4  Netherlands            7.403  10.942           0.930           71.550   

   Freedom  Generosity  Corruption  
0    0.961      -0.019       0.182  
1    0.934       0.134       0.196  
2    0.936       0.211       0.668  
3    0.809      -0.023       0.708  
4    0.887       0.213       0.379  
```

In [10]:

```python
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"
print("Original columns:", df.columns.tolist())

df_subset = df.iloc[:, :9]  # Select only the first 9 columns
df_subset.columns = ['Country','Region','Happiness_Score','GDP','Social_Support',
                     'Life_Expectancy','Freedom','Generosity','Corruption']

print(f"Dataset: {len(df_subset)} countries, {len(df_subset.columns)} columns")
print(df_subset.head())
```

```
Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'uppe
rwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expect
ancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder
score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'E
xplained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Expl
ained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']
Dataset: 137 countries, 9 columns
       Country  Region  Happiness_Score    GDP  Social_Support  \
0      Finland   7.804            0.036  7.875           7.733
1      Denmark   7.586            0.041  7.667           7.506
2      Iceland   7.530            0.049  7.625           7.434
3       Israel   7.473            0.032  7.535           7.411
4  Netherlands   7.403            0.029  7.460           7.346

   Life_Expectancy  Freedom  Generosity  Corruption
0           10.792    0.969      71.150       0.961
1           10.962    0.954      71.250       0.934
2           10.896    0.983      72.050       0.936
3           10.639    0.943      72.697       0.809
4           10.942    0.930      71.550       0.887
```

In [11]:

```python
region_map = {
    'Finland': 'Western Europe',
    'Denmark': 'Western Europe',
    'Iceland': 'Western Europe',
    'Netherlands': 'Western Europe',
    'Sweden': 'Western Europe',
    'Norway': 'Western Europe',
    'Germany': 'Western Europe',
    'France': 'Western Europe',
    'Spain': 'Western Europe',

    'India': 'South Asia',
    'Pakistan': 'South Asia',
    'Bangladesh': 'South Asia',

    'China': 'East Asia',
    'Japan': 'East Asia',
    'South Korea': 'East Asia',

    'Nigeria': 'Sub-Saharan Africa',
    'Kenya': 'Sub-Saharan Africa',
    'Ethiopia': 'Sub-Saharan Africa',
    'Congo (Kinshasa)': 'Sub-Saharan Africa',
    'Sierra Leone': 'Sub-Saharan Africa',

    'Brazil': 'Latin America',
    'Argentina': 'Latin America',
    'Mexico': 'Latin America'
}

df_clean['Region'] = df_clean['Country'].map(region_map)
df_clean = df_clean.dropna(subset=['Region'])
```

In [7]:

```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default = 'browser'
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"
df_clean = df[['Country name', 'Ladder score']].copy()
df_clean.columns = ['Country', 'Happiness_Score']
region_map = {
    'Finland': 'Western Europe',
    'Denmark': 'Western Europe',
    'Iceland': 'Western Europe',
    'Netherlands': 'Western Europe',
    'Sweden': 'Western Europe',
    'Norway': 'Western Europe',
    'Germany': 'Western Europe',
    'France': 'Western Europe',
    'Spain': 'Western Europe',

    'India': 'South Asia',
    'Pakistan': 'South Asia',
    'Bangladesh': 'South Asia',

    'China': 'East Asia',
    'Japan': 'East Asia',
    'South Korea': 'East Asia',

    'Nigeria': 'Sub-Saharan Africa',
    'Kenya': 'Sub-Saharan Africa',
    'Ethiopia': 'Sub-Saharan Africa',
    'Congo (Kinshasa)': 'Sub-Saharan Africa',
    'Sierra Leone': 'Sub-Saharan Africa',

    'Brazil': 'Latin America',
    'Argentina': 'Latin America',
    'Mexico': 'Latin America'
}
df_clean['Region'] = df_clean['Country'].map(region_map)
df_clean['Region'] = df_clean['Region'].fillna('Other')
print("Columns:", df_clean.columns)
print(df_clean.head())
print(df_clean['Region'].value_counts())
region_avg = df_clean.groupby('Region')['Happiness_Score'].mean().reset_index()
region_avg = region_avg.sort_values('Happiness_Score')

print(region_avg)  # debug
fig1 = px.bar(
    region_avg,
    x='Happiness_Score',
    y='Region',
    orientation='h',
    color='Region'
)

fig1.update_xaxes(range=[0, region_avg['Happiness_Score'].max()])

fig1.update_layout(
    title="Western Europe Shows the Highest Happiness Levels Among Selected Regions",
    plot_bgcolor='white'
)
fig1.show()
```

```
Columns: Index(['Country', 'Happiness_Score', 'Region'], dtype='object')
       Country  Happiness_Score          Region
0      Finland            7.804  Western Europe
1      Denmark            7.586  Western Europe
2      Iceland            7.530  Western Europe
3       Israel            7.473           Other
4  Netherlands            7.403  Western Europe
Other                 114
Western Europe          9
Sub-Saharan Africa      5
Latin America           3
East Asia               3
South Asia              3
Name: Region, dtype: int64
               Region  Happiness_Score
4  Sub-Saharan Africa         3.980800
3          South Asia         4.291000
2               Other         5.480491
0           East Asia         5.966000
1       Latin America         6.159667
5      Western Europe         7.224667
```

In [15]:

```
pip install plotly
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: plotly in c:\programdata\anaconda3\lib\site-packages (5.9.
0)
Requirement already satisfied: tenacity>=6.2.0 in c:\programdata\anaconda3\lib\site-packag
es (from plotly) (8.0.1)
Note: you may need to restart the kernel to use updated packages.
```

In [11]:

```python

import pandas as pd
import plotly.express as px
import plotly.io as pio

pio.renderers.default = 'browser'   # or 'notebook' if you prefer


df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"

df_clean = df[[
    'Country name',
    'Ladder score'
]].copy()

df_clean.columns = ['Country', 'Happiness_Score']

print("Data preview:")
print(df_clean.head())


top8 = df_clean.nlargest(8, 'Happiness_Score').copy()
top8['Group'] = 'Top 8'

bottom8 = df_clean.nsmallest(8, 'Happiness_Score').copy()
bottom8['Group'] = 'Bottom 8'


combined = pd.concat([bottom8, top8])
combined = combined.sort_values('Happiness_Score')


global_avg = df_clean['Happiness_Score'].mean()


fig = px.bar(
    combined,
    x='Happiness_Score',
    y='Country',
    orientation='h',
    color='Group',
    color_discrete_map={
        'Top 8': 'red',
        'Bottom 8': 'blue'
    }
)


fig.add_vline(x=global_avg, line_dash="dash")


fig.update_xaxes(range=[0, combined['Happiness_Score'].max()])

fig.update_layout(
    title="Top Countries Are Nearly Twice as Happy as the Lowest Ranked Ones",
    plot_bgcolor='white'
)


fig.show()
```

```
Data preview:
       Country  Happiness_Score
0      Finland            7.804
1      Denmark            7.586
2      Iceland            7.530
3       Israel            7.473
4  Netherlands            7.403
```

In [10]:

```python
import pandas as pd
import plotly.graph_objects as go
import plotly.io as pio
pio.renderers.default = 'browser'
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"
df_clean = df[[
    'Country name',
    'Ladder score',
    'Logged GDP per capita',
    'Freedom to make life choices'
]].copy()
df_clean.columns = [
    'Country',
    'Happiness_Score',
    'GDP',
    'Freedom'
]
region_map = {
    'Finland': 'Western Europe',
    'Denmark': 'Western Europe',
    'Iceland': 'Western Europe',
    'Netherlands': 'Western Europe',
    'Sweden': 'Western Europe',
    'Norway': 'Western Europe',
    'Germany': 'Western Europe',
    'France': 'Western Europe',
    'Spain': 'Western Europe',

    'India': 'South Asia',
    'Pakistan': 'South Asia',
    'Bangladesh': 'South Asia',

    'China': 'East Asia',
    'Japan': 'East Asia',
    'South Korea': 'East Asia',

    'Nigeria': 'Sub-Saharan Africa',
    'Kenya': 'Sub-Saharan Africa',
    'Ethiopia': 'Sub-Saharan Africa',
    'Congo (Kinshasa)': 'Sub-Saharan Africa',
    'Sierra Leone': 'Sub-Saharan Africa',

    'Brazil': 'Latin America',
    'Argentina': 'Latin America',
    'Mexico': 'Latin America'
}
df_clean['Region'] = df_clean['Country'].map(region_map)
df_clean['Region'] = df_clean['Region'].fillna('Other')
regions = [
    'Western Europe',
    'Latin America',
    'East Asia',
    'Sub-Saharan Africa',
    'South Asia'
]
subset = df_clean[df_clean['Region'].isin(regions)]
grouped = subset.groupby('Region')[['GDP', 'Freedom']].mean().reset_index()
print(grouped)
fig3 = go.Figure()
fig3.add_trace(go.Bar(
    x=grouped['Region'],
    y=grouped['GDP'],
    name='GDP'
))
fig3.add_trace(go.Bar(
    x=grouped['Region'],
    y=grouped['Freedom'],
    name='Freedom'
))
fig3.update_layout(
```

```python
    barmode='group',
    title="Western Europe Leads in Both Economic Strength and Personal Freedom Compared
    xaxis_title="Region",
    yaxis_title="Average Value",
    plot_bgcolor='white'
)

fig3.show()
```

```
              Region        GDP   Freedom
0          East Asia  10.349000  0.799333
1      Latin America   9.797000  0.826667
2         South Asia   8.661333  0.836000
3 Sub-Saharan Africa   7.818800  0.691600
4     Western Europe  10.853667  0.895556
```

In [ ]:

```python

```